

MAT111 Assignment 2

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Problem 1

Let $n = 2m + 1$ for n odd. Then $n^2 = (2m + 1)^2 = 4m^2 + 4m + 1 = 2(2m^2 + 2m) + 1$ which is another odd number, so n^2 is odd when n is odd.

Problem 2

Consider the number 24. It is not divisible by 48 but it is divisible by 6 and 8.

Problem 3

Let the two irrational numbers be x, y .

If question asks for $x, y \in \mathbb{R}$: $\pi + (-\pi) = 0 \in \mathbb{Q}$

If question asks for $x, y \in \mathbb{R}^+$: Let $x = \sqrt{2}$ and $y = 2 - \sqrt{2}$. $x + y = 2$ is rational while x is irrational by common knowledge. Also, we know that the sum of two rational numbers is always rational ($\frac{p_1}{q_1} + \frac{p_2}{q_2} = \frac{p_1 q_2 + p_2 q_1}{q_1 q_2}$). But $2 - y = x$ is irrational so y must be irrational too.

Problem 4

Consider $f(x) = x^2$ which is a continuous function. But $f(x) = f(-x)$ and $x \neq -x$ for $x \neq 0$.

Problem 5

Consider $f(x) = 0$. Then $f(x + y) = f(x) + f(y) = 0 + 0 = 0$ but clearly $f(x) \neq x$.

Problem 6

$x \nmid yz \implies x \nmid y \wedge x \nmid z \iff x \mid y \vee x \mid z \implies x \mid yz$. Without loss of generality, let $x \mid y$. Then $\frac{y}{x} \in \mathbb{Z}$. Since integer multiplied by an integer is an integer, $\frac{y}{x}z \in \mathbb{Z}$ as well.

Problem 7

For $n \in \mathbb{N}$, $2^n - 1$ is prime $\implies n$ is prime $\iff n$ is composite $\implies 2^n - 1$ is composite.

If n is composite, it can be written as $n = pq$ where $p, q > 0$. Then $2^n - 1 = 2^{pq} - 1$. Writing this as a geometric series sum yields

$$\begin{aligned} 2^{pq} - 1 &= \underbrace{1 + \dots + 2^{p-1}}_{2^p - 1} + \dots + 2^{pq-2} \\ &= 2^p - 1 + 2^p \underbrace{(1 + \dots + 2^{p-1})}_{2^p - 1} + \dots + 2^{pq-1} \\ &\vdots \\ &= 2^p - 1 + 2^p(2^p - 1) + 2^{2p}(2^p - 1) + \dots + 2^{pq-p}(2^p - 1) \end{aligned}$$

By above algebra, we see that $2^p - 1 \mid 2^{pq} - 1$.

Problem 8

By contrapositive, $\gcd(a, n) \neq \gcd(b, n) \implies a \not\equiv b \pmod{n}$ is the statement we want to prove.

This implies $\exists d > 1 (d \mid a) \wedge (d \mid n) \wedge (d \nmid b)$. Since $a \equiv 0 \pmod{d}$ and $b \not\equiv 0 \pmod{d}$, $a + (-b) \not\equiv 0 \pmod{d}$. But every multiple of n must also be a multiple of d . Since d does not divide $a - b$, neither does n .

Problem 9

As it is currently written, the statement "If $x \mid yz$, then $x \mid y$ and $x \mid z$ " is false.

Problem 10

By contrapositive, x is odd and y is even $\implies x^2(y+3)$ is odd is the statement we want to prove.

Letting $x = 2n + 1$ and $y = 2m$, then $x^2(y+3) = (2n+1)^2(2m+3) = 8n^2m + 8nm + 2m + 12n^2 + 12n + 2 + 1$. Factoring this expression gives $2(4n^2m + 4nm + 2m + 6n^2 + 6n + 1) + 1$ which is of the form of an odd number.

Problem 11

By contrapositive, $a = 2n \implies 4 \mid a^2$ is the statement we want to prove.

$a^2 = (2n)^2 = 4n^2 = 4(n^2)$ is divisible by 4.

Problem 12

By contrapositive, $x < 0 \implies x^5 + 7x^3 + 5x < x^4 + x^2 + 8$ is the statement we want to prove.

Since $x < 0$, $x^{2n+1} < 0$ while $x^{2n} > 0$ for $x \in \mathbb{R} \setminus \{0\}$. Hence $x^5 < 0$, $x^3 < 0$, $x < 0$ and $x^4 > 0$, $x^2 > 0$, $8 > 0$. So left hand side is less than 0 and right hand side is greater than 0, meaning left hand side is less than right hand side.

Problem 13

By contrapositive, if $f(x) \neq f(y)$ then $f(x)$ is not a constant function.

Problem 14

Assume the largest even number is n . A larger even number is $2n$. Hence we have reached a contradiction.

Problem 15

Assume $A \cap (B - A) \neq \emptyset$. But $B - A$ does not contain any element of A so there cannot be any common elements in A and $B - A$.

Problem 16

Assume the sum of an even integer ($2n$) and an odd integer ($2m - 1$) is even ($2n + 2m - 1 = 2(n + m) - 1 = 2k$). But this is a contradiction.

Problem 17

Suppose P is such a statement. No logical operators ($\neg$, $\wedge$, $\vee$, etc) accept such a statement. Therefore P is not inside the system of formal logic I have been using. Hence I cannot answer this question.

Problem 18

Assume n is odd and n^2 is even. Then $n = 2m + 1$ and $n^2 = (2m + 1)^2 = 2(2m^2 + 2m) + 1$ which is odd which is a contradiction.

Problem 19

Assume such a pair exists. Let $a = 2^x a'$ and $b = 2^y b'$ where a', b' are odd. Then the equation becomes $(2^x a')^2 = 2(2^y b')^2 \implies 2^{2x} a'^2 = 2^{2y+1} b'^2$. But the power of 2 differs in left and right hand side so we have reached a contradiction.

Problem 20

Assume $\sqrt[3]{2} = \frac{p}{q}$. Then $p^3 = 2q^3$. Let $p = 2^x p'$ and $q = 2^y q'$ where p', q' are odd. Then the equation becomes $(2^x p')^3 = 2(2^y q')^3 \implies 2^{3x} p'^3 = 2^{3y+1} q'^3$. But the power of 2 differs in left and right hand side so we have reached a contradiction.

Problem 21

Assume $\sqrt{6} = \frac{p}{q}$. Then $p^2 = 6q^2$. Let $p = 2^x 3^u p'$ and $q = 2^y 3^v q'$ where $\gcd(p', 6) = 1$ and $\gcd(q', 6) = 1$. Then the equation becomes $(2^x 3^u p')^2 = 2 \cdot 3 \cdot (2^y 3^v q')^2 \implies 2^{2x} 3^{2u} p'^2 = 2^{2y+1} 3^{2v+1} q'^2$. But the power of 2 and 3 differs in left and right hand side so we have reached a contradiction.

Problem 22

Assume there exists a rational point on the curve. Letting $x = \frac{p_1}{q_1}$ and $y = \frac{p_2}{q_2}$, we have

$$\begin{aligned} \left(\frac{p_1}{q_1}\right)^2 + \left(\frac{p_2}{q_2}\right)^2 &= 3 \\ \implies (p_1 q_2)^2 + (p_2 q_1)^2 &= 3(q_1 q_2)^2 \end{aligned}$$

Now, let $k = \gcd(q_1, q_2)$. Then, let $q_1 = km_1$ and $q_2 = km_2$ which gives

$$\begin{aligned} (p_1 km_2)^2 + (p_2 km_1)^2 &= 3(k^2 m_1 m_2)^2 \\ \implies (p_1 m_2)^2 + (p_2 m_1)^2 &= 3(k m_1 m_2)^2 \end{aligned}$$

Now, let $a = p_1 m_2, b = p_2 m_1, c = km_1 m_2$ and consider $\gcd(a, b, c)$. Considering $\gcd(b, c) = \gcd(p_2 m_1, km_1 m_2)$, observe that $\gcd(p_2, km_2) = 1$ and $\gcd(p_2, m_1) \mid m_1$. Also, $\gcd(m_1, m_2) = 1$. Hence $\gcd(b, c) = \gcd(m_1, km_1) = m_1$. Therefore we have $\gcd(a, b, c) = \gcd(a, \gcd(b, c)) = \gcd(p_1 m_2, m_1) = 1$.

Now, taking the equation modulo 3

$$\begin{aligned} a^2 + b^2 &= 3c^2 \\ \implies a^2 + b^2 &\equiv 0 \pmod{3} \end{aligned}$$

By checking $(a, b) \in \{0, 1, 2\} \times \{0, 1, 2\}$, we find that $3 \mid a^2$ and $3 \mid b^2$. So $3 \mid a$ and $3 \mid b$. Writing $a = 3x$ and $b = 3y$

$$\begin{aligned} 3c^2 &= (3x)^2 + (3y)^2 \\ \implies 3c^2 &= 9(x^2 + y^2) \\ \implies c^2 &= 3(x^2 + y^2) \\ \implies c^2 &\equiv 0 \pmod{3} \end{aligned}$$

So we find that $3 \mid c^2 \implies 3 \mid c$ but we know that $\gcd(a, b, c) = 1$, hence we have reached a contradiction.

To show that $\sqrt{3}$ is irrational, consider the equation $x^2 + y^2 = 3$ with $x = 0$. Since there are no rational points on this curve at all, there are no rational points with $x = 0$ either.

Problem 23

$g : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z} \times \mathbb{Z}, g(m, n) = (m + n, m + 2n)$

Injectivity

Suppose $g(m_1, n_1) = g(m_2, n_2)$. Then $(m_1 + n_1, m_1 + 2n_1) = (m_2 + n_2, m_2 + 2n_2)$. Solving this system for m_1 and n_1 , we find that $m_1 = m_2$ and $n_1 = n_2$. Therefore, by the contrapositive statement of $(m_1, n_1) \neq (m_2, n_2) \implies g(m_1, n_1) \neq g(m_2, n_2)$, which is $g(m_1, n_1) = g(m_2, n_2) \implies (m_1, n_1) = (m_2, n_2)$, we have found that g is injective.

Surjectivity

Suppose $g(m, n) = (a, b)$. Then $(m + n, m + 2n) = (a, b)$. Solving this system for m and n gives $m = 2a - b$ and $n = b - a$ and clearly $(2a - b, b - a) \in \mathbb{Z} \times \mathbb{Z}$ so g is surjective.

Problem 24

$$\theta : \{0, 1\} \times \mathbb{N} \rightarrow \mathbb{Z}, \theta(a, b) = a - 2ab + b$$

Injectivity

Suppose $\theta(a, b) = n$. Choosing $a = 0$, we must have $b = n$. On the other hand, choosing $a = 1$, we must have $b = \frac{n-a}{1-2a}$. But $1 - 2a = -1$, so for $b \in \mathbb{N}$, we must have $n - a < 0 \implies n < a \implies n \leq 0$. This means $\theta(a, b) \in \mathbb{N} \implies a = 0$ in which case there is a unique b , and also $\theta(a, b) \in \mathbb{Z} \setminus \mathbb{N} \implies a = 1$ in which case there is also a unique b . Therefore θ is injective.

Surjectivity

For $\theta(a, b) = n \in \mathbb{N}$, $(a, b) = (0, n)$. For $\theta(a, b) = n \in \mathbb{Z} \setminus \mathbb{N}$, $(a, b) = (-1, a - n)$. So θ is surjective.

Since θ is both injective and surjective, θ is bijective.

Problem 25

$$\theta : \{0, 1\} \times \mathbb{N} \rightarrow \mathbb{Z}, \theta(a, b) = (-1)^a b$$

Injectivity

Firstly, $\theta(a, b) > 0 \iff a = 0$ and $\theta(a, b) < 0 \iff a = 1$.

Then, consider $\theta(0, b_1) = \theta(1, b_2)$. This is only possible if $b_2 = -b_1$ but then either $b_1 \in \mathbb{N}$ or $b_2 \in \mathbb{N}$, not both. Hence $\theta(a_1, b_1) \neq \theta(a_2, b_2)$ if $a_1 \neq a_2$.

If $a_1 = a_2 = a$, $\theta(a, b_1) = \theta(a, b_2) \implies (-1)^a b_1 = (-1)^a b_2 \implies b_1 = b_2$ so there is only 1 input for each output. So θ is injective.

Surjectivity

$\nexists (a, b) \in \{0, 1\} \times \mathbb{N}, \theta(a, b) = 0$. So θ is not surjective.

Since θ is not both injective and surjective, it is not bijective.

Problem 26

$$f : \mathbb{N} \rightarrow \mathbb{Z}, f(n) = \frac{(-1)^n(2n-1)+1}{4}$$

Injectivity

$$\begin{aligned} f(x) &= f(y) \\ \implies \frac{(-1)^x(2x-1)+1}{4} &= \frac{(-1)^y(2y-1)+1}{4} \\ \implies (-1)^x(2x-1) &= (-1)^y(2y-1) \\ \implies |2x-1| &= |2y-1| \end{aligned}$$

But $x, y \in \mathbb{N} \implies 2x - 1, 2y - 1 \in \mathbb{N}$ so we have $2x - 1 = 2y - 1 \implies x = y$. Hence f is injective.

Surjective

$$\begin{aligned} f(x) &= n \\ \implies (-1)^x(2x - 1) &= 4n - 1 \end{aligned}$$

If $n \leq 0$, then x must be odd. Letting $x = 2y - 1$

$$\begin{aligned} \implies -(2(2y - 1) - 1) &= 4n - 1 \\ \implies 4y - 2 - 1 &= 1 - 4n \\ \implies y &= \frac{4 - 4n}{4} \\ \implies y &= 1 - n \\ \implies x &= 1 - 2n \geq 1 \end{aligned}$$

On the other hand, if $n > 0$, then x must be even. Letting $x = 2y$ and similar algebra yields $x = n \geq 1$. Hence f is surjective.

Since f is both injective and surjective, it is bijective.

Problem 27

$$f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}, f(m, n) = 2^{m-1}(2n - 1)$$

Injectivity

$$\begin{aligned} f(m_1, n_1) &= f(m_2, n_2) \\ \implies 2^{m_1-1}(2n_1 - 1) &= 2^{m_2-1}(2n_2 - 1) \\ \implies 2^{m_1-m_2}(2n_1 - 1) &= 2n_2 - 1 \end{aligned}$$

Clearly, right hand side is odd, and left hand side is odd only if $m_1 = m_2$. Then $2^0(2n_1 - 1) = 2n_2 - 1 \implies n_1 = n_2$ so f is injective.

Surjectivity

Since every natural number can be factorized into its even part and odd part, f is surjective.

Since f is both injective and surjective, it is bijective.

Problem 28

Let A_n be any set of size n .

Base case

$$\mathcal{P}(A_1) = \{\emptyset, A_1\} = 2 = 2^1.$$

Inductive step

Suppose we construct A_n from A_{n-1} like so: $A_n := A_{n-1} \cup \{x\}$. Then

$$\mathcal{P}(A_n) = \mathcal{P}(A_{n-1}) \cup \bigcup_{S \in \mathcal{P}(A_{n-1})} \{S \cup \{x\}\}$$

Therefore, $\mathcal{P}(A_n) = 2\mathcal{P}(A_{n-1}) = 2 \cdot 2^{n-1} = 2^n$.

Problem 29

Base case

For $n = 1$, $1 + x \leq (1 + x)^1$.

Inductive step

We need to show that if $1 + nx \leq (1 + x)^n$ then $1 + (n + 1)x \leq (1 + x)^{n+1}$.

But consider $(1 + nx)(1 + x) - (1 + nx + x) = 1 + nx + x + nx^2 - 1 - nx - x = nx^2 \geq 1$.

Therefore, $(1 + nx + x) \leq (1 + nx)(1 + x) \implies (1 + nx + x) \leq (1 + x)^n(1 + x)$. Equivalently $(1 + (n + 1)x) \leq (1 + x)^{n+1}$.

Problem 30

We can construct the sequence a inductively.

Base case

a_1 can be any natural number. Then we get a_1^2 which is a perfect square.

Inductive step

We know that $a_1^2 + a_2^2 + \dots + a_n^2 = S_n^2$. Now we need to generate a_{n+1} such that $S_n^2 + a_{n+1}^2 = S_{n+1}^2$. But we also know that $\sum_{k=1}^m 2k - 1 = m^2$. Therefore, we may write

$$\begin{aligned} S_n^2 + a_{n+1}^2 &= S_{n+1}^2 \\ \implies S_n^2 + \sum_{k=1}^{a_{n+1}} (2k - 1) &= \sum_{k=1}^{S_{n+1}} (2k - 1) \end{aligned}$$

Thinking wishfully, we would rather be the right hand side a single term instead of a series. Letting $a_{n+1} + 1 = S_{n+1}$ gets the desired result:

$$\begin{aligned} S_n^2 &= \sum_{k=1}^{a_{n+1}+1} (2k - 1) - \sum_{k=1}^{a_{n+1}} (2k - 1) \\ \implies S_n^2 &= 2a_{n+1} + 1 \end{aligned}$$

Now, we require S_n to always be odd so that we may construct a_{n+1} . To achieve this, we use induction again.

Base case

We actually need to adjust the original base case. Instead of any natural number, a_1 must be any odd number.

Inductive step

We know that S_n is odd. Therefore, $S_n^2 = (2k - 1)^2 = 4k^2 - 4k + 1 = 2a_{n+1} + 1$. Then $a_{n+1} = 2(k^2 - k)$ which is even. So we get $S_{n+1}^2 = S_n^2 + a_{n+1}^2 = \text{odd} + \text{even} = \text{odd}$.

Since we are able to generate a_n for $n \in \mathbb{N}$, we can generate a valid sequence.

Problem 31

Base case

$$0 = 0 \cdot 2^0.$$

$$1 = 1 \cdot 2^0.$$

Inductive step

We have the binary representation $a_{k-1}a_{k-2}\dots a_1a_0$ for all $n < 2^k$. For $2^k \leq m < 2^{k+1}$, we may write $m = 1 \cdot 2^k + n$. This way, we get the binary representation for m as $1a_{k-1}\dots a_0$.

Problem 32

Base case

By De Morgan's theorem, $(A_1 \cap A_2)' = A_1' \cup A_2'$.

Also by De Morgan's theorem, $(A_1 \cup A_2)' = A_1' \cap A_2'$.

Inductive step

Since intersection is associative, we can write $(A_1 \cap A_2 \cap \dots \cap A_n \cap A_{n+1})'$ as $((A_1 \cap A_2 \cap \dots \cap A_n) \cap A_{n+1})'$.

Let $B = (A_1 \cap A_2 \cap \dots \cap A_n)$. By De Morgan's theorem, $(B \cap A_{n+1})' = B' \cup A_{n+1}'$. But we know that $B' = A_1' \cup \dots \cup A_n'$.

Part 2 can be done similarly.